

Applicant's Name:

Yucheng Bao

Proposed Topic/Title of Research:

**Long-Context Inference with Long-Term Memory: Cost-Bounded
Context Compression, Retrieval and State Reuse**

Abstract

Large language models are increasingly expected to operate over long-running tasks: multi-session assistants, evolving code repositories, longitudinal medical records, legal case histories, and enterprise knowledge bases. These settings cannot be handled reliably by simply placing more text into the prompt. Longer contexts increase cost and latency, and they do not by themselves solve evidence selection, temporal change, stale summaries, privacy boundaries, or error accumulation across sessions. This proposal studies long-context inference as a memory management problem under explicit cost and reliability constraints. The proposed research will design a framework that represents long-term memory with provenance, versioning, uncertainty, and access control; compresses context according to token and latency budgets; retrieves evidence in a time-aware and contradiction-aware manner; and reuses intermediate state only when validity conditions are satisfied. The expected outcome is a principled approach to memory-augmented inference that improves the trade-off among answer quality, faithfulness, latency, and operational cost.

Key Words

Long-context inference; long-term memory; retrieval-augmented generation; context compression; state reuse; faithful reasoning; cost-bounded inference

1. Research Background and Significance

Recent progress in large language models has made it possible to build assistants that can write code, analyze documents, answer complex questions, and support decision-making. However, many important applications are not single-turn tasks. They unfold across days, months, or years, and the relevant information is distributed across documents, conversations, tool outputs, database records, and prior decisions. A coding assistant may need to reason about an evolving repository and remember design choices made in previous sessions. A legal assistant may need to connect current questions with prior filings, contracts, regulations, and exceptions. A medical assistant may need to interpret a patient's history across visits, diagnoses, tests, and treatments. These tasks require continuity, but the information needed for

continuity cannot be treated as an unbounded prompt.

The direct solution, extending the context window or concatenating more documents, is insufficient. It increases token cost, latency, memory consumption, and energy use. It can also reduce reliability, because relevant evidence may be hidden among irrelevant passages, outdated material may be treated as current, and the model may fail to use information located in the middle of a long context. Retrieval-augmented generation (RAG) addresses part of the problem by selecting external evidence at query time, but standard RAG often assumes relatively static documents and short-lived interactions. It does not fully address multi-session memory, temporal versioning, long-horizon dependencies, or the computational cost of repeated inference.

Long-term memory methods provide another direction. Work such as LongMem shows that language models can benefit from external memory mechanisms that preserve past information beyond a single context window. More broadly, memory-augmented language models and agent systems have explored storing user preferences, summaries, facts, and interaction traces. These methods are promising, but memory also creates new failure modes. A summary may omit a detail that becomes crucial later. An abstractive memory may contain unsupported claims. Retrieved evidence may be semantically similar but temporally invalid. Cached computation may be reused even after the document, model, prompt, or access permission has changed. In high-stakes settings, these errors are not merely inconvenient; they undermine trust and auditability.

This proposal starts from the view that the core challenge is not simply "how to fit more tokens", but how to decide what information should be retained, compressed, retrieved, verified, and reused under limited resources. A reliable long-context system should maintain memory as a controlled store rather than an informal collection of summaries. Each memory item should have provenance, freshness, version information, uncertainty, and access scope. Retrieval should respect time, source authority, and contradiction risk. Compression should reduce cost while preserving verifiable links to original evidence. State reuse should be treated as conditional computation, valid only when the assumptions behind the cached state still hold.

The significance of this research lies in integrating three lines of work that are usually studied separately: context compression, retrieval, and inference-state reuse. Compression reduces prompt length but can damage faithfulness. Retrieval improves grounding but can increase latency and introduce stale or conflicting evidence. State reuse reduces repeated computation but can create correctness and security risks. A practical long-context system must coordinate these mechanisms through a shared policy that balances cost, latency, evidence quality, and safety. This research aims to provide such a policy framework and evaluate it empirically.

2. Current State of Domestic and International Research

Existing research can be grouped into four related areas.

First, long-context model architectures extend the amount of text a model can process through improved attention mechanisms, positional encoding, sparse attention, recurrence, or memory tokens. Benchmarks such as LongBench have helped evaluate long-context understanding across tasks, and studies such as "Lost in the Middle" show that models do not always use long contexts reliably. These findings suggest that longer windows are useful but not a complete solution: when context becomes very large or changes over time, the system still needs selection, organization, and verification.

Second, retrieval-augmented generation connects language models with external knowledge sources. Classic RAG retrieves relevant documents and conditions generation on them. Later work improves retrieval through dense-sparse hybrid search, reranking, query rewriting, self-reflection, and citation-based verification. These methods provide stronger grounding, but typical RAG pipelines are still often query-local: they retrieve for the current question without maintaining a durable model of previous sessions, temporal changes, or accumulated evidence.

Third, long-term memory systems attempt to preserve information across interactions. Some systems store summaries, user profiles, facts, vector embeddings, or episodic traces. They are useful for personalization and continuity, but many designs lack formal treatment of provenance, versioning, uncertainty, and retention. This makes it difficult to determine whether a memory item should be trusted, updated, forgotten, or overridden.

Fourth, efficient inference systems reduce serving cost through KV cache management, prefix reuse, paged memory, cache compression, and attention-state reuse. Systems such as vLLM demonstrate the importance of memory management for serving throughput, while recent work on prompt caching and KV cache compression shows that repeated context can be exploited to reduce latency and bandwidth. However, computation reuse in a long-term memory setting raises additional validity questions: a cached state is only correct under specific model, prompt, document, and permission conditions.

The research gap is therefore an integrated one. Current methods may optimize for longer context, better retrieval, smaller memory, or faster inference, but few provide a unified treatment of cost-bounded memory management for long-context reasoning. This project will address that gap by studying how compression, retrieval, and state reuse can be jointly controlled under explicit constraints on cost, freshness, faithfulness, and auditability.

3. Personal Research Background

My academic background provides preparation for this research, but the proposed project is not a direct continuation of any single previous project. I received my BSc (Hons) in Computing from The Hong Kong Polytechnic University with First Class Honours and a minor in Applied Mathematics, and I am currently pursuing an MSc in Computer Science at The Chinese University of Hong Kong. My training covers algorithms, machine learning,

deep learning, information retrieval, data processing, and computer systems.

Several experiences have shaped my ability to carry out this project. I have worked on retrieval-augmented question answering over domain-specific documents, which helped me understand the importance of document parsing, hybrid retrieval, and evidence grounding. I have also studied KV cache compression for LLM inference, which gave me practical awareness of memory cost, latency, and quality trade-offs. My current work on LLM serving simulation and profiling further strengthens my ability to evaluate inference systems beyond model accuracy alone. These experiences are relevant because the proposed research requires both algorithmic design and careful system-level evaluation. They will inform the implementation and experimentation, while the research questions themselves are defined by the broader scientific problem of reliable long-context inference.

4. Research Questions and Objectives

This project will investigate the following research questions:

1. How should a long-context system decide what to store as raw evidence, structured memory, extractive compression, or abstractive summary under token, latency, and storage budgets?
2. How can memory representations preserve provenance, versioning, uncertainty, and access scope so that generated answers remain auditable?
3. How can retrieval be made aware of time, document version, source authority, and contradiction in evolving corpora?
4. Under what conditions can retrieved results, tool outputs, prefix states, or attention/KV states be safely reused without compromising correctness, privacy, or robustness?
5. How can a controller choose among retrieval, compression, state reuse, direct generation, and refusal when evidence is insufficient?

The main objective is to develop a cost-bounded long-context inference framework that improves the Pareto trade-off among answer quality, faithfulness, latency, and cost. A second objective is to design evaluation protocols for long-term memory systems, including stress tests for temporal change, contradictory evidence, stale memory, and delayed dependencies.

5. Proposed Methodology

The proposed framework contains five connected components.

5.1 Managed Long-Term Memory Store

The first component is a managed memory store that treats memory items as auditable objects. Each item will include content, representation type, source identifier, source span, timestamp, document version, confidence or uncertainty score, access namespace, and retention rule. The memory store will support multiple levels, such as session-level memory, project-level memory, and corpus-level memory. It will contain both lossless evidence and compressed representations. Lossless snippets are required for high-stakes claims; structured facts support efficient filtering and reasoning; summaries support broad context but must link back to source evidence.

The research problem here is to determine what metadata and policies are necessary for reliable use of memory. I will compare memory schemas with different levels of provenance and versioning, and evaluate how these choices affect citation correctness, update handling, and user-facing answer quality.

5.2 Cost-Bounded Context Compression

The second component is a compression pipeline governed by explicit budgets. Instead of treating summarization as the default compression method, the system will maintain several forms of compression:

Extractive compression preserves important source spans directly.

Structured compression converts information into typed records, such as entities, events, decisions, constraints, API changes, obligations, or temporal relations.

Abstractive compression produces natural-language summaries, but only with source links and uncertainty indicators.

The project will study when each representation is appropriate. For example, legal clauses, code changes, and medical measurements may require extractive or structured preservation, while exploratory background information may tolerate abstractive summarization. The system will also study compression scheduling: when to compress immediately, when to delay compression until repeated access occurs, and when to discard information because its expected future value is low.

5.3 Hierarchical and Time-Aware Retrieval

The third component is retrieval over multi-scale and evolving information. The system will build hierarchical indexes, such as chunk-file-repository for code, clause-document-case for legal analysis, or visit-episode-patient for medical-style records. A query planner will first infer the likely scope and time range, then use hybrid sparse and dense retrieval. Retrieved candidates will be reranked by relevance, recency, provenance quality, source authority, and contradiction risk.

A key part of this component is contradiction-aware retrieval. In evolving corpora, the most semantically similar item may not be the correct one if it has been superseded. The system should detect when older and newer evidence conflict, expose the conflict to the generator, and prefer answer forms that acknowledge version-dependent conclusions.

5.4 Safe State Reuse

The fourth component is safe reuse of intermediate computation. Long-running applications often repeat the same documents, prefixes, retrieval calls, and tool operations. Reusing these states can reduce latency and cost, but reuse must be conditional. The cache key should encode the model version, tokenizer, prompt template, system instruction, document hash, tool version, access namespace, and timestamp. Invalidation should occur when the underlying document changes, the model changes, the prompt contract changes, or permission boundaries change.

This component will compare several reuse targets: retrieval results, reranking results, tool outputs, prefix states, and attention/KV states. The evaluation will measure not only cache hit rate but also end-to-end latency, memory footprint, stale reuse rate, and answer correctness under document updates.

5.5 Cost-Faithfulness Controller

The final component is a controller that selects an inference strategy for each query. Possible actions include retrieving raw evidence, using structured memory, using summaries, expanding more context, reusing cached state, generating directly, or refusing to answer. The controller will optimize a multi-objective function involving token cost, latency, evidence coverage, freshness, uncertainty, and risk level.

The controller can begin as a rule-based policy and later be trained with offline logs or bandit-style adaptation. Hard constraints will be used for safety: for example, an answer that makes factual claims must include supporting evidence, and claims unsupported by retrieved evidence should be revised or refused. This design separates efficiency optimization from minimum reliability requirements.

6. Experimental Design and Evaluation

The experiments will focus on long-horizon and evolving information settings. Candidate tasks include multi-session codebase question answering over changing repositories, document question answering over versioned policy or legal corpora, and longitudinal record analysis where later questions depend on earlier events. These tasks are suitable because they require memory, temporal reasoning, and evidence selection rather than only short-context semantic matching.

Baselines will include full-context prompting when feasible, standard RAG, summary-only

memory, retrieval-only memory, hierarchical retrieval without memory policies, and caching without strict validity checks. Ablation studies will remove individual components such as provenance metadata, structured compression, temporal retrieval, contradiction detection, or cache invalidation.

Evaluation will include quality, faithfulness, and cost metrics. Quality metrics will include task accuracy, answer helpfulness, evidence recall, and robustness to updates. Faithfulness metrics will include citation correctness, entailment between answer claims and evidence, contradiction rate, and inappropriate refusal rate. Cost metrics will include input tokens, output tokens, latency, throughput, cache hit rate, memory footprint, compression ratio, and cost per query.

Special stress tests will be designed for long-term memory failure modes:

Stale memory: older evidence conflicts with newer evidence.

Missing dependency: a small detail from an early session is needed later.

Compression distortion: a summary changes or omits a critical fact.

Permission change: previously accessible evidence should no longer be used.

Repeated context: state reuse should reduce cost without changing answers.

The expected result is a set of trade-off curves rather than a single absolute score. The framework should show when additional memory improves accuracy, when compression begins to damage faithfulness, when retrieval depth becomes inefficient, and when state reuse is beneficial or unsafe.

7. Challenges and Risk Mitigation

The first challenge is factual distortion from compression. This will be mitigated by preserving source links, preferring extractive or structured memory for high-risk facts, and using entailment checks before summaries are used as support.

The second challenge is retrieval over heterogeneous and evolving corpora. This will be addressed through hierarchical indexing, metadata filters, temporal constraints, version tracking, and contradiction-aware reranking.

The third challenge is unsafe state reuse. Strict cache keys, time-to-live policies, document hashing, tenant or namespace isolation, and invalidation tests will reduce stale or unauthorized reuse.

The fourth challenge is evaluation. Long-term memory errors may appear only after multiple updates or sessions. The project will therefore construct controlled benchmark scenarios that intentionally introduce temporal changes, conflicts, and delayed dependencies.

The fifth challenge is over-optimizing for cost at the expense of reliability. The controller will

use hard evidence requirements so that cost-saving actions cannot bypass minimum faithfulness constraints.

8. Expected Contributions

This research is expected to make four contributions.

First, it will propose a unified framework for long-context inference that treats memory, compression, retrieval, and state reuse as coordinated decisions under explicit cost and reliability constraints.

Second, it will develop memory representations and policies that preserve provenance, versioning, uncertainty, retention, and access scope.

Third, it will design and evaluate a cost-faithfulness controller that chooses inference strategies according to query needs, budget, freshness, and evidence requirements.

Fourth, it will provide empirical evaluation protocols and stress tests for long-term memory systems, helping compare systems not only by accuracy but also by latency, cost, faithfulness, and auditability.

In practical terms, this research can support more reliable assistants for software engineering, legal and policy analysis, healthcare-style records, and enterprise knowledge work. In academic terms, it aims to clarify how long-term memory should be represented, controlled, and evaluated in language model systems.

9. Research Plan

In the first stage, I will conduct a systematic literature review on long-context models, retrieval-augmented generation, memory-augmented language models, context compression, inference caching, and KV/state reuse. I will define the target task settings, memory schema, and evaluation metrics.

In the second stage, I will implement the managed memory store, compression pipeline, and hierarchical retrieval baseline. This stage will focus on provenance, versioning, temporal filtering, and evidence-linked compression.

In the third stage, I will add safe state reuse and the first version of the controller. I will evaluate the system under token budgets, latency targets, document updates, and controlled contradiction scenarios.

In the fourth stage, I will refine the controller, conduct robustness and safety experiments, compare against baselines, and prepare publications based on the findings.

References

- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W., Rocktaschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems*, 33.
- Wang, W., Dong, L., Cheng, H., Liu, X., Yan, X., Gao, J., & Wei, F. (2023). Augmenting Language Models with Long-Term Memory. *Advances in Neural Information Processing Systems*, 36, 74530-74543.
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics*, 12, 157-173.
- Bai, Y., Lv, X., Zhang, J., Lyu, H., Tang, J., Huang, Z., Du, Z., Liu, X., Zeng, A., Hou, L., Dong, Y., Tang, J., & Li, J. (2024). LongBench: A Bilingual, Multitask Benchmark for Long Context Understanding. *Proceedings of ACL 2024*.
- Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J. E., Zhang, H., & Stoica, I. (2023). Efficient Memory Management for Large Language Model Serving with PagedAttention. *Proceedings of the ACM Symposium on Operating Systems Principles*.
- Gim, I., Chen, G., Lee, S., Sarda, N., Khandelwal, A., & Zhong, L. (2024). Prompt Cache: Modular Attention Reuse for Low-Latency Inference. *Proceedings of MLSys 2024*.
- Liu, S., Sheng, Y., Zhou, C., Ruwase, O., Mai, L., Zaharia, M., & Stoica, I. (2024). CacheGen: Fast Context Loading for Language Model Applications. *Proceedings of ACM SIGCOMM 2024*.
- Jiang, H., Wu, Q., Luo, C., Li, D., Lin, C. Y., Yang, Y., & Qiu, L. (2023). LLMingua: Compressing Prompts for Accelerated Inference of Large Language Models. *Proceedings of EMNLP 2023*.
- Xu, F., Shi, W., & Choi, E. (2024). RECOMP: Improving Retrieval-Augmented LMs with Compression and Selective Augmentation. *Proceedings of ICLR 2024*.
- Asai, A., Wu, Z., Wang, Y., Sil, A., & Hajishirzi, H. (2024). Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *Proceedings of ICLR 2024*.